

SRS Rough Draft

Functional Requirements

FR-01 | GUI

The system shall provide a GUI where users can interact with UVSim.

FR-02 | Program Loading

The system shall allow users to enter a BasicML program into the application

FR-03 | Program Loading

The system shall load BasicML program into memory

FR-04 | Instruction Validation

The system shall detect instruction containing invalid operation codes

FR-05 | Code Editor

The GUI shall provide an area for users to enter or view BasicML code

FR-06 | Program Loading Control

The GUI shall provide a control for loading a BasicML program

FR-07 | Program Execution

The GUI shall provide a control for executing the loaded BasicML program

FR-08 | Accumulator Display

The GUI shall display the current value of the accumulator

FR-09 | Instruction Display

The GUI shall display the current instruction being executed

FR-10 | Program Output

The GUI shall display program output to the user

FR-11 | Program Input

The GUI shall provide a way for the user to enter values from read instructions.

FR-12 | Execution Status

The GUI shall indicate when program execution has completed

FR-13 | Error Display

The GUI shall display an error message when an error occurs

FR-14 | Step Execution

The system shall allow the user to execute a program one line at a time

FR-15 | Execution Continuation

The system shall allow the user to continue execution after stepping through instructions

Non-functional requirements

NFR-01 | Resource Usage

The simulator shall run within reasonable CPU and memory usage on a typical computer

NFR-02 | Recoverability

The system shall allow the user to reset the simulator to a known initial state after an error or completed execution

NFR-03 | Responsiveness

The system shall begin executing a BasicML program within 2 seconds of user selecting 'execute' option